Project Overview

Olist E-commerce Data Analysis

Analysis of the Olist E-commerce Data Analysis

- The analysis of the Olist e-commerce dataset followed standard data analytics best practices and leveraged nine interconnected datasets: orders, products, sellers, customers, geolocation, order payments, order items, order reviews, and product categories
- This section summarizes the workflow, key findings, insights, and final recommendations derived from the analysis.

Olist E-commerce Dataset

- Dataset: 9 linked datasets (Orders, Products, Sellers, Customers, Payments, Reviews, etc.
- Period: 2016 – 2018
- Tool Used: Power BI & Microsoft Excel
- Objective: Analyze business performance, customer behavior, logistics, and product insights

Data & Methodology(workflow)

- Problem Definition / Business Questions: Clear analytical objectives were defined, including evaluating business performance, identifying top-performing product categories, and understanding factors influencing customer review scores.
- Data Collection & Integration: Nine separate CSV datasets were collected and properly modeled using relational keys to create a unified dataset suitable for comprehensive analysis.
- Data Cleaning & Preprocessing: Missing values, outliers, and structural inconsistencies were handled appropriately. Data formats were standardized, and feature engineering was performed where necessary (e.g., delivery time, order value) using Microsoft Excel & Power BI.
- Exploratory Data Analysis (EDA) & Modeling: Statistical summaries and visualizations—such as trend charts, heatmaps, and geospatial analysis—were used to identify patterns and relationships within the data.
- Validation & Interpretation: Findings were validated and interpreted to ensure accuracy and relevance, enabling the extraction of actionable business insights.
- Communication & Visualization: Results were communicated to stakeholders through structured reports and interactive dashboards developed using Power BI.

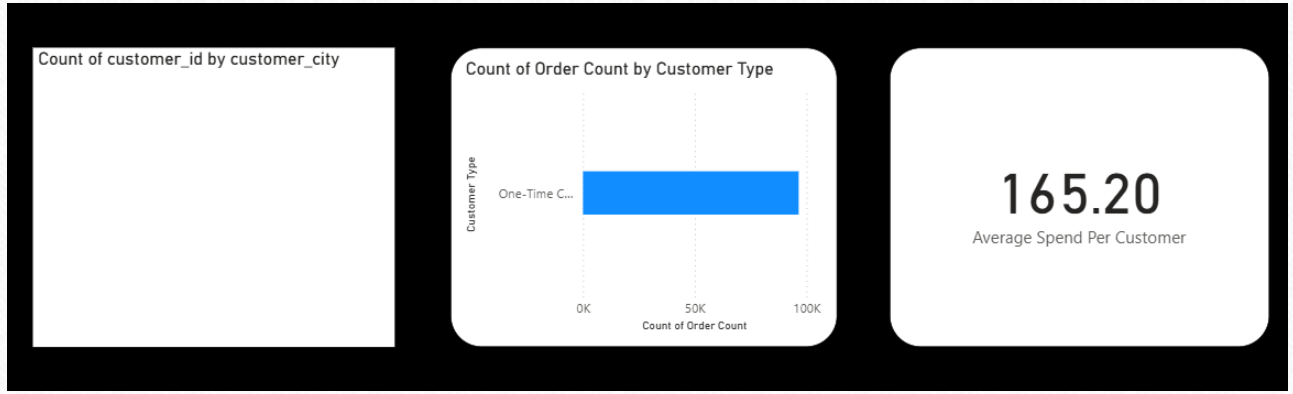
Findings and Insights

- Order and Revenue Trends: Olist processed approximately 100,000 orders between 2016 and 2018, generating total revenue of about ₦16 million. The business experienced steady growth from September 2016 to September 2018, with peak order volumes typically occurring around March.



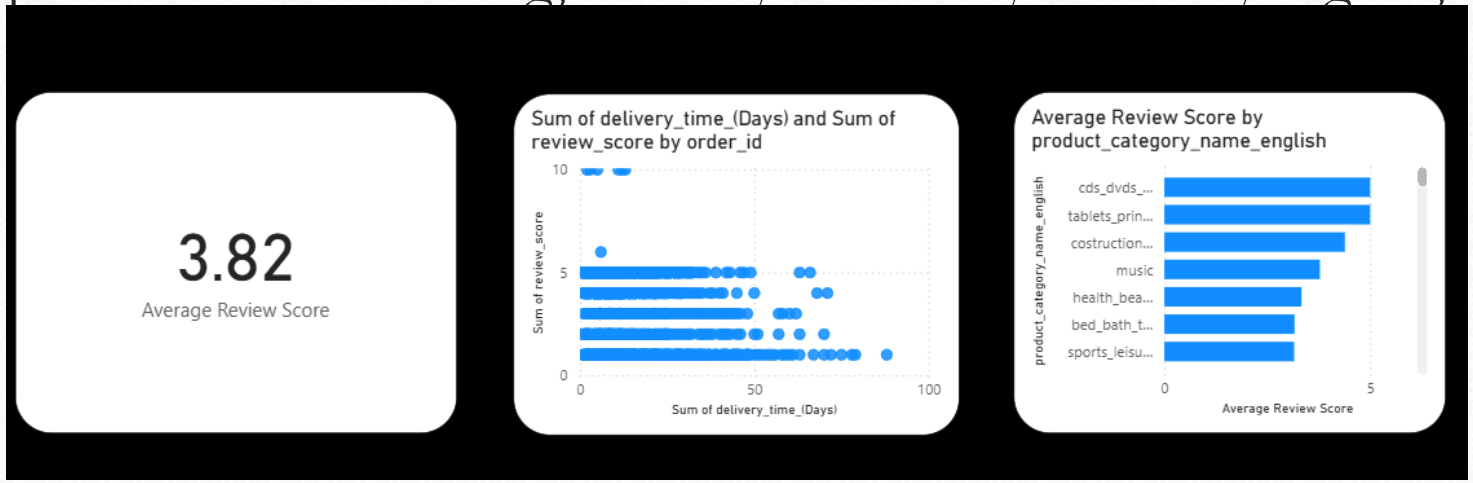
Customer Satisfaction Insights

- The average product rating is approximately 4.09 out of 5, and the order cancellation rate is low (around 0.6%), indicating generally high customer satisfaction. However, a noticeable proportion of 1-star reviews suggests quality or service issues in specific product categories.



Logistics & Delivery Performance

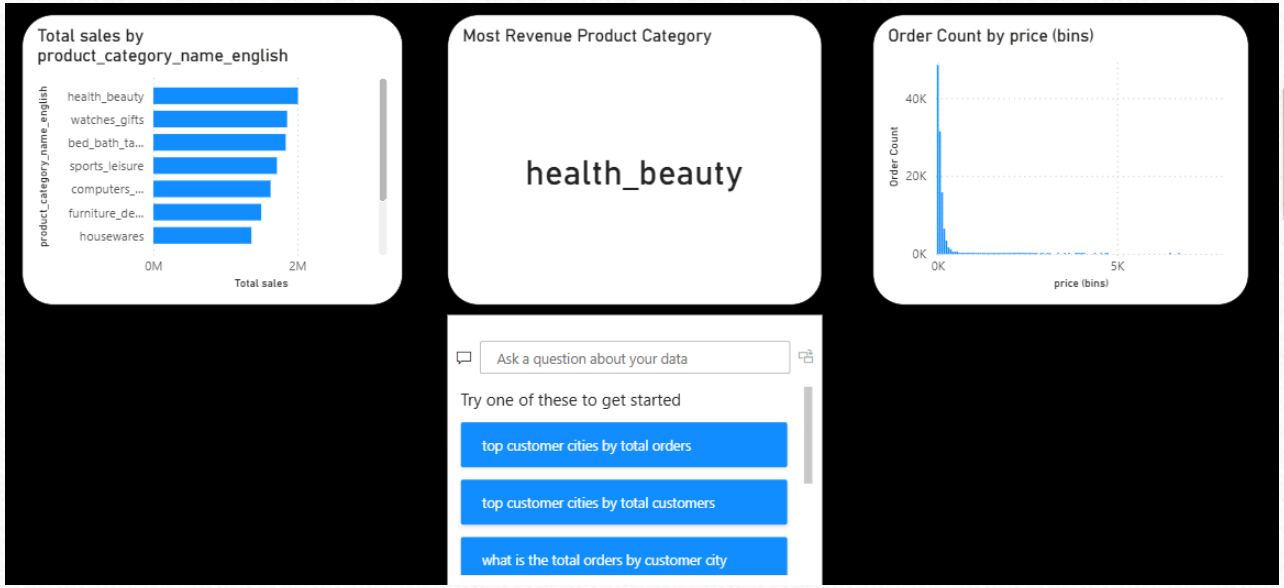
- Delivery time has a strong influence on customer review scores, with faster deliveries correlating with higher satisfaction. While overall logistics performance is strong, delivery efficiency varies by region, and certain



Product Performance

- Top categories by order volume:

1. Health & Beauty
2. Bed & Bath
3. Watches & Gifts



High-margin categories include:

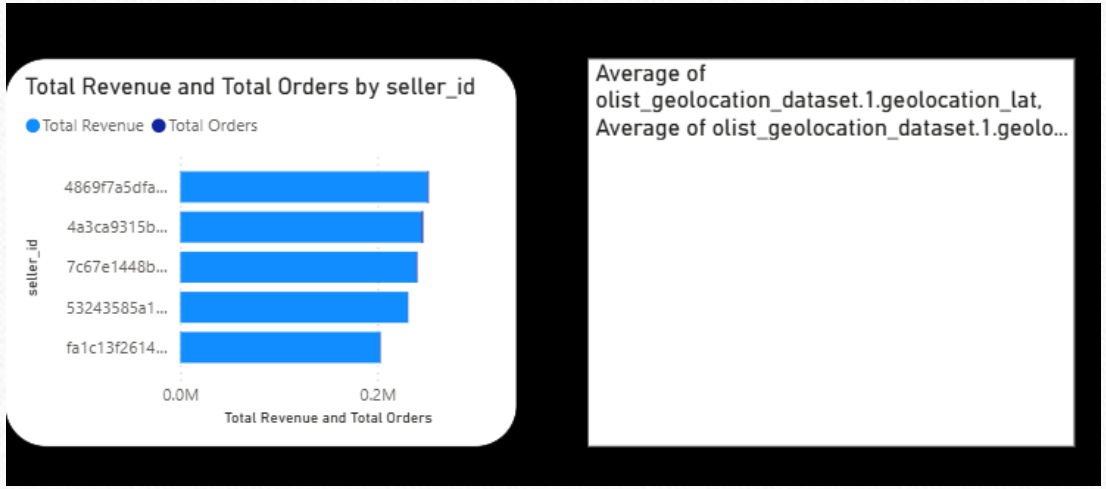
- Telefonia_fixa
- Drinks

Customer Behavior & Payments

- Credit cards are the dominant payment method, used by approximately 76–78% of customers. A major challenge identified is the low customer retention rate, with a churn rate of roughly 87.5%, indicating that most customers make only one purchase.

Geolocation Insight

- São Paulo has the highest concentration of customers, making it a critical region for targeted marketing and logistics optimization.



Key Recommendation

- Improve quality control for low-rated categories
- Optimize logistics in underperforming delivery routes
- Implement loyalty programs to boost repeat purchases
- Personalize marketing using customer segmentation
- Expand high-margin products and onboard more sellers

Final Recommendations

Based on the insights obtained, the following recommendations are proposed to enhance Olist's growth and customer loyalty:

- Enhance Product Quality Control: Investigate the product categories receiving low (1 star) reviews and implement quality control measures to address consistency and meet customer expectations.
- Optimize Logistics: Analyze and address the specific bottlenecks and delays in high priority or low-performing shipping routes to improve overall delivery times and, consequently, review scores.
- Improve Customer Retention: Focus on strategies to convert first-time buyers into repeat customers. Implement loyalty programs with exclusive discounts or free shipping for targeted high-density customer locations like Sao Paulo.
- Personalize Marketing: Utilize customer segmentation data to tailor recommendations and offers, creating a more engaging and personalized shopping experience for different customer groups. •
- Strategic Expansion: Capitalize on high-profit-margin products (e.g., telefonia_fixa, drinks) through targeted marketing, and consider strategies to onboard more sellers to improve product variety and balance the distribution of seller locations.